CP(M) 2026 Invigilated MiniZinc Assessment, mock paper: sample answers at about band C

This document gives a set of answers to the mock paper at about band C, written as a student might write them in the assessment. The answers contain deliberate gaps and errors, which the comment at the end identifies and relates to the band criteria. Line numbers refer to the appendices of the paper.

Question 1.1. (D) starts everything unburnt. (E) works out which vertices the fire spreads to in each turn, from the vertices that were burning at the previous turn and their neighbours. (F) then adds the vertex that is chosen at that turn. So `burnt` follows the burning process turn by turn, and `burnt[t, v]` is true when v is burning at the end of turn t.

Question 1.2. No. The second disjunct is there so that a vertex can still be chosen when every vertex has already been reached by the fire. Without it, the model cannot choose a vertex at a turn where the whole graph is already burning, so it may need more turns. For example, on `path-9` the modified model would give 4 instead of 3.

Question 2.1. `y[0]` to `y[3]` are 1 and the rest are 0, and `turns` $= 4$. `burn_turn[4]` $= 1$, `burn_turn[10]` $= 2$, `burn_turn[14]` $= 3$, `burn_turn[16]` $= 4$, and all the others are 0, so the sequence printed is `[4, 10, 14, 16]`. For $u = 7$, vertex 4 has radius 3 and is at distance 3 from 7, so the sum in (E) is at least 1. Vertex 13 is similar.

Question 2.2. `turns` times `x[v, r]` is non-linear because it multiplies two variables. M has to be at least as big as n, and `UB` $\leq n$, so it is big enough.

Question 3, first part. In Appendix A, change line 44 to `constraint forall(t in 1..UB)(count(v in 1..n)(burn_turn[v] = t) <= k);` so that up to k vertices can be chosen in each turn. In Appendix B, change line 53 to `constraint forall(r in RADIUS)(sum(v in 1..n)(x[v, r]) <= k * y[r]);`. The rest of the models stay the same.

Question 3, second part. Choose 3 and 8 at turn 1 and 12 and 15 at turn 2. I tried to find a way to burn `path-16` in two turns with two vertices per turn and could not, so two turns are not enough.

Comment. The meaning of the variables and constraints is explained correctly, and the routine parts are mostly correct: the values in Question 2.1, the nonlinearity in Question 2.2, the sequence in Question 3 and the change to the ILP model. The arguments about the consequences of modelling choices are incomplete or partly wrong.

- **Question 1.1** describes the constraints but is not a proof.
- **Question 1.2** has the right idea but an unchecked example: the modified model still gives 3 on `path-9`. A graph such as K_2, on which it fails, was needed.
- **Question 2.1** does not verify (E) for $u = 13$.
- **Question 2.2** gets M wrong: what matters is that $M \geq$ `turns`, and comparing with n does not explain why.
- **Question 3, first part:** the modified CP model keeps (C), which forces exactly `turns`

chosen vertices in total. It therefore gives 4 turns instead of 3 on `path-16`.

- **Question 3, second part:** failing to find a two-turn sequence is not an argument that none exists.